

SCAM: Scan Channel Attention Mamba-Based Network for Remote Sensing Change Detection

Junyi Zhang , Renwen Chen , Fei Liu , Aocheng He, Boyu Zheng, and Hao Liu 

Abstract—Remote sensing change detection (RSCD) aims to detect changes on the Earth's surface over time by analyzing remote sensing imagery taken at two different time points for the same location. Current change detection methods based on convolutional neural networks (CNNs) and transformer struggle to balance global modeling capability with computational complexity. Recently, Mamba has garnered substantial interest in the RSCD domain due to its global receptive field and linear computational complexity. However, existing Mamba-based methods are constrained by fixed scanning strategies, limiting their ability to adaptively capture spatial contextual information and enhance local features. In this article, we propose scan channel adaptive mamba (SCAM), a novel framework designed to dynamically capture and enhance key features. Specifically, we introduce a channel adaptive state-space scan strategy, which employs a lightweight scorer to select critical scanning channels and reinforce local features of the patched feature maps. Furthermore, a difference fusion module is introduced to integrate multiscale difference information with the global features extracted from bitemporal images. In addition, we design a lightweight, CNN-based multiscale change decoder to aggregate multiscale change features and generate the final change mask. Experimental results on four benchmark datasets—LEVIR-CD, WHU-CD, SYSU-CD, and CLCD, demonstrate that the proposed SCAM achieves F1 scores of 91.01%, 93.14%, 83.60%, and 78.84%, respectively, outperforming the other comparative methods.

Index Terms—Channel adaptive state-space scan (CASS), Mamba, remote sensing change detection (RSCD), state-space model (SSM).

I. INTRODUCTION

REMOTE sensing change detection (RSCD) aims to detect changes on the Earth's surface over time by analyzing remote sensing imagery taken at two different time points for the same location. It has seen broad adoption across multiple fields,

including urban development [1], [2], [3], land use evaluation [4], environmental surveillance [5], [6], [7], [8], and disaster impact analysis [9], [10]. Among the available data sources, optical high-resolution RS images are particularly prominent in RSCD due to their ability to capture fine-grained surface information. Traditional RSCD approaches, such as transformation-based [11], [12], [13]; classification-based [14], [15]; and object-based [16], [17] methods, rely solely on shallow image features, making them ineffective in handling complex change regions.

In recent years, the advancement of convolutional neural networks (CNNs) has introduced numerous novel solutions to the RSCD domain [18], [19], [20], [21]. CNN-based methods localize regions of interest where changes have occurred by generating segmentation masks, achieving promising results across various RSCD datasets. However, the intrinsic locality of convolution operations constrains the receptive field of CNNs. Although some approaches attempt to mitigate this limitation by employing deeper architectures, dilated convolutions, or attention mechanisms, their capacity for global modeling remains restricted. The introduction of vision transformer has effectively mitigated the receptive field constraints associated with CNNs [22], [23], [24], [25]. The transformer framework has shown impressive results in CD tasks [26], [27], [28], [29], [30], [31], primarily due to its capacity to model spatial relationships and capture global context via self-attention mechanisms. However, its application to image-based tasks introduces a challenge of quadratic computational complexity, which substantially increases the processing cost. While some approaches attempt to enhance efficiency by restricting the attention window's size or stride, these strategies often come at the cost of a reduced receptive field.

Consequently, one of the central challenges in RSCD remains the effective modeling of spatiotemporal context related to change, while keeping computational demands low. In recent studies, the state-space model (SSM) [32] has demonstrated notable effectiveness in processing long-sequence data. Building on this, Mamba [33] introduces time-dependent parameters, significantly enhancing the model's adaptability and achieving notable success in computer vision [34], [35], [36], [37], [38], [39], [40], [41]. With its linear computational complexity, Mamba offers a compelling alternative to Transformer-based architectures by preserving strong global modeling capabilities while substantially reducing computational overhead. However, the 2D-selective-scan (SS2D) strategy adopted in VMamba [35] focuses primarily on global feature modeling, potentially neglecting important local information. To mitigate this issue,

Received 7 August 2025; revised 8 September 2025; accepted 4 December 2025. Date of publication 8 December 2025; date of current version 22 December 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 51675265, in part by Zhiyuan Laboratory under Grant ZYL2024020, in part by the Interdisciplinary Innovation Fund for Doctoral Students of Nanjing University of Aeronautics and Astronautics under Grant KXKCXJJ202201, and in part by the Advantage Discipline Construction Project Funding of University in Jiangsu Province under Grant PAPD. (Corresponding author: Junyi Zhang.)

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Digital Object Identifier 10.1109/JSTARS.2025.3641390

LocalMamba [42] incorporates local context by partitioning images into patches and processing each patch independently. RS-Mamba [43] proposes an omnidirectional-selective-scan (OSS) strategy designed specifically for RS imagery, enabling the integration of spatial context in multiple directions. CD-Lamba [44] introduces a multiscale cross-temporal locally adaptive state-space scan (CT-LASS) strategy to effectively highlight changes and refine changes' representations feature generation. Despite their effectiveness, these methods struggle to focus on local features within critical scanning channels, and cause varying degrees of disruption to boundary continuity during the modeling process.

To overcome the limitations discussed above, we introduce a novel network architecture named scan channel adaptive mamba (SCAM). Within the encoder, channel adaptive state-space scan (CASS) block uses a lightweight scorer to adaptively select key scanning channels and enhance local features of the patched feature maps, thereby improving the network's sensitivity to critical features. Meanwhile, it adopts a novel S-shaped scanning path to improve the boundary continuity of change targets. In addition, a difference fusion module (DFM) is introduced to extract bitemporal difference features across multiple spatial scales, which further strengthens the model's capability to detect subtle changes. To infer the final change mask, we design a multiscale change decoder (MCD) that directly processes the multiscale difference features obtained from DFM. In conclusion, the main contributions of this work can be summarized as follows.

- 1) We propose a novel SCAM and design a CASS strategy, which adaptively enhances local features of bitemporal images while maintaining global perception capability.
- 2) We introduce a DFM to incorporate multiscale difference information into bitemporal image features, thereby enhancing the network's capacity to detect subtle changes.
- 3) We design an MCD to integrate multiscale change features and predict the change map.
- 4) Comprehensive experiments on four publicly available RSCD datasets confirm that the proposed SCAM model delivers notable performance gains.

In this article, Section II introduces the related work in RSCD field. Section III describes the proposed modules and strategies. Section IV presents the results of comparative experiments and ablation studies. Section V summarizes the work of this article.

II. RELATED WORKS

Over the past few years, driven by the advancements in deep learning and the extensive availability of high-resolution RS imagery, traditional approaches that primarily depend on shallow feature extraction have encountered challenges in achieving the precision required for RSCD tasks. Consequently, deep learning-based methods for RSCD have gained considerable attention and become a focal point of research, which can be broadly divided into three categories: CNN-based, transformer-based, and Mamba-based methods.

A. CNN-Based Method

The advancement of CNNs and their strong capability in feature extraction have facilitated their widespread adoption in

RSCD. Daudt et al. [18] introduced FC-Siam-Conc, FC-EF, and FC-Siam-Diff, marking the first use of CNNs in RSCD. Among them, FC-Siam-Conc achieves fusion by concatenating bitemporal images as a unified input, and it performs the best within the FC-series models. With the emergence of powerful backbone architectures, such as VGG [45] and ResNet [46], deeper convolutional layers can capture richer image features, thereby improving detection accuracy and further advancing CNN-based methods. Fang et al. [21] introduced SNUNet, a model that leverages densely connected convolutional structures to effectively capture deep spatiotemporal dependencies. Han et al. [47] developed hierarchical attention network, a discriminative Siamese network that fuses multiscale information while enhancing fine-grained features.

Although the aforementioned methods have demonstrated promising performance, they remain limited in their ability to enhance difference information and suppress irrelevant change noise. Consequently, numerous studies have concentrated on leveraging difference information. In W-shaped hierarchical network (WNet) [48], the difference enhancement module (DEM) is introduced, which efficiently strengthens and learns bitemporal difference features through direct differencing and concating operations. Jiang et al. [49] proposed a multidimensional attention-aggregation network, in which a difference guidance module was designed. Its three submodules are dedicated to learning multiscale difference features and enhancing critical information while suppressing irrelevant information from both spatial and channel perspectives. Huang et al. [50] proposed a spatiotemporal enhancement and interlevel fusion network (SEIFNet), which effectively addresses the challenges of pseudo-change interference and insufficient utilization of multiscale features by integrating spatio-temporal difference enhancement with an adaptive context fusion strategy.

While CNN-based methods perform well in RSCD, their limited receptive fields hinder global feature extraction, making it difficult to model long-range dependencies and leading to weaker detection results.

B. Transformer-Based Method

Recently, transformer excels at modeling sequential data and addresses the limitations of CNNs, prompting their widespread adoption in RSCD research.

Chen et al. [28] proposed BIT, which models spatiotemporal context by extracting bitemporal features using a weight-shared ResNet-18. BIT converts multitemporal images into semantic tokens and captures spatiotemporal dependencies through a token-based framework. Liu et al. [29] introduced MSCANet, which attains a balanced tradeoff between accuracy and efficiency through a multiscale context-aware module and a spatial-channel attention mechanism. Bandara and Patel [27] proposed ChangeFormer, composed of a transformer-based encoder and a multilayer perception (MLP) decoder, which leverages attention mechanisms to model and handle long-range dependencies. Zhang et al. [26] presented SwinSUNet, a model tailored for RSCD tasks based on the Swin-transformer and U-Net architectures. It integrates the global modeling capacity of the Swin-Transformer with the multiscale feature fusion strengths of

U-Net. Li et al. [51] designed TransUNetCD, a hybrid architecture that combines the global context modeling of transformers with the U-Net framework. A differential enhancement module is introduced to suppress redundant features and enhance feature quality.

Despite their impressive performance in RSCD, Transformer-based approaches often struggle with high computational demands, as their quadratic complexity in image processing leads to substantial resource consumption, which can hinder practical deployment in resource-constrained environments.

C. Mamba-Based Method

SSM, originally inspired by modern control theory, models the relationship between input stimuli and system responses using continuous-time differential equations. Mamba incorporates a selection mechanism within the SSM framework to address the challenges Transformer face in handling long sequence data, particularly with respect to efficiency and the ability to model long-range dependencies. VMamba combines SSM with visual tasks, achieving global receptive field while maintaining linear complexity, offering a novel solution for RSCD tasks.

Currently, Mamba-based methods have been widely adopted in the RSCD field. Chen et al. [52] first introduced Mamba into bitemporal RS image analysis, demonstrating strong performance in binary change detection, semantic change detection, and building damage assessment. Building on this, Zhang et al. [53] proposed CDMamba, which enhances detailed change detection in dense prediction tasks by integrating global and local features via the SRCM block. Zhao et al. [43] introduced RS-Mamba, which employs the OSS strategy to scan from multiple directions, thereby improving global feature extraction. Wu et al. [44] proposed CD-Lamba, integrating a novel local adaptive enhancement strategy into Mamba-based RSCD tasks, thereby substantially enhancing the model's local perception capacity.

Although Mamba-based methods have shown encouraging performance in RSCD tasks, there is still room for improvement in their scanning strategies and feature extraction processes. In this work, we present a novel scanning strategy to enhance local feature representation. We also introduce difference fusion module and multiscale change decoder to extract and process multiscale fused features.

III. METHODOLOGY

A. Preliminaries

The SSM offers a structured approach for characterizing the generation of time series data by employing state-space representations, assuming linear time-invariant (LTI) systems, and applying mapping functions. In this framework, a hidden state $h(t) \in \mathbb{R}$ serves as an intermediate variable that transforms the one-dimensional input function or sequence $x(t)$ into the output $y(t) \in \mathbb{R}$. This relationship is typically formulated using a system of linear ordinary differential equations as shown in the following:

$$h'(t) = Ah(t) + Bx(t) \quad (1)$$

$$y(t) = Ch(t) \quad (2)$$

where $h'(t)$ denotes the temporal derivative of $h(t)$, the matrix $A \in \mathbb{R}^{N \times N}$ characterizes the system's evolution parameters. The matrices $B \in \mathbb{R}^{N \times 1}$ and $C \in \mathbb{R}^{1 \times N}$ serve as input and output projection operators, respectively.

In the field of deep learning, data representations, such as vectors and matrices are typically discrete. To enable the incorporation of continuous-time systems into deep learning models, SSM applies a zero-order hold to convert the continuous parameters A and B into their discrete representations $\bar{A}$ and $\bar{B}$. This transformation introduces a time scaling factor $\Delta \in \mathbb{R}^D$, defined as follows:

$$\bar{A} = \exp(\Delta A) \quad (3)$$

$$\bar{B} = (\Delta A)^{-1} (\exp(\Delta A) - I) (\Delta B). \quad (4)$$

Therefore, the discrete forms of (1) and (2) can be represented as follows:

$$h(t) = \bar{A} h_{t-1} + \bar{B} x_t \quad (5)$$

$$y(t) = Ch(t). \quad (6)$$

Nevertheless, the SSM framework, grounded in the LTI assumption, inherently restricts the input-output relationship to remain constant over time. To address this limitation, recently introduced models (i.e., Mamba or S6) extend the traditional SSM by allowing key parameters $\bar{B}, C, \Delta$ to vary in response to the input sequence. This enhancement enables the model to perform selective information processing across time steps, dynamically modulating contextual representations. Furthermore, Mamba adopts hardware-efficient algorithms to optimize GPU memory layout, effectively reducing memory usage and computational costs.

B. Overall Architecture

The proposed SCAM structure is shown in Fig. 1. It takes as input a pair of bitemporal RS images, $T_1 \in \mathbb{R}^{H \times W \times 3}$ and $T_2 \in \mathbb{R}^{H \times W \times 3}$, which are 3-channel RGB images with a resolution of $H \times W$. First, the CASS encoder extracts features from the bitemporal images through four stages.

Stage I partitions the original image via patch embedding, whereas each of the subsequent three stages begins with a downsampling operation on the feature maps. In each stage, the encoder employs the CASS Block to perform global feature modeling and to enhance local features within patches along key scanning channels. Next, the DFM integrates the features extracted at each stage from bitemporal images. This module not only enhances global feature representation but also fuses the differential features between the bitemporal images, generating multiscale fused features. Finally, the multiscale features generated by DFM are fed into a lightweight MCD, which produces the final change mask through multilevel upsampling and fusion operations followed by a classifier.

C. Channel Adaptive State-Space Scan Encoder

Existing Mamba-based scanning strategies struggle to maintain local features within the patches. To address this issue,

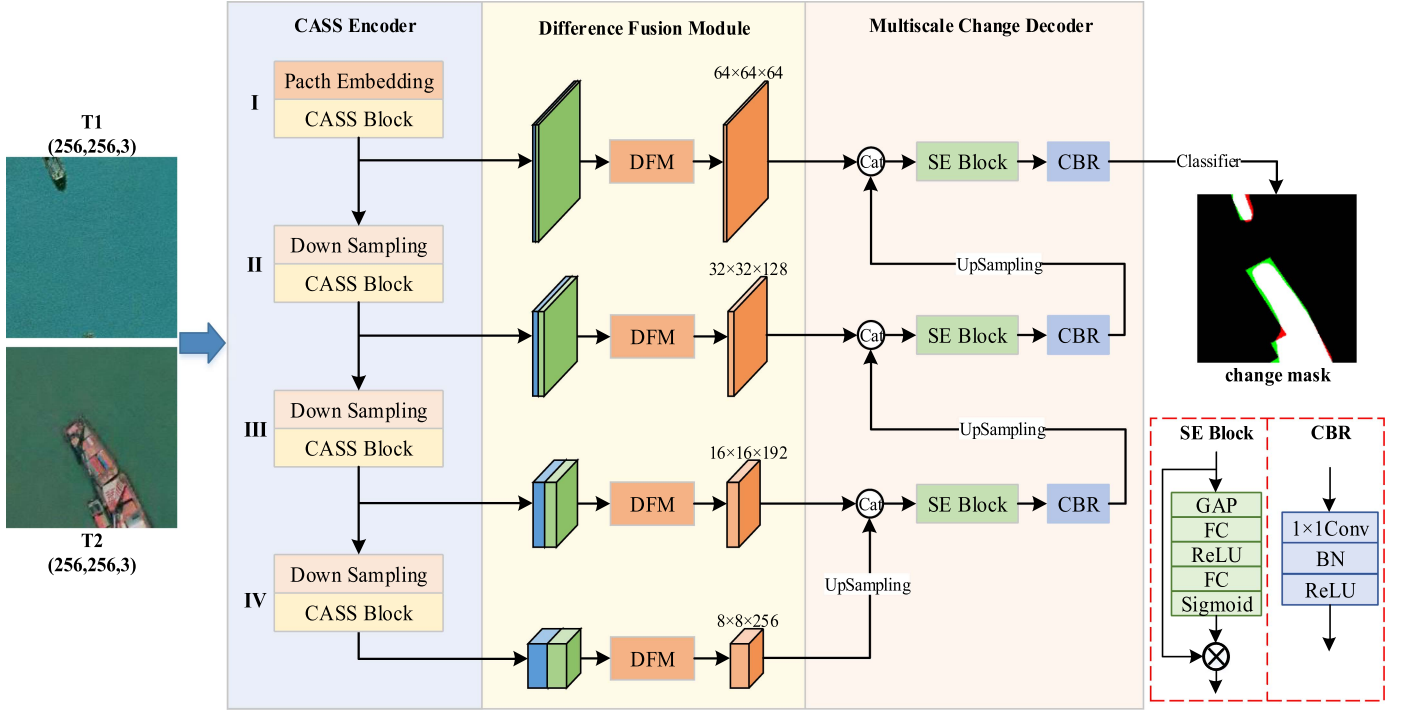


Fig. 1. Overall architecture of our proposed SCAM includes four stages of CASS encoder, four difference fusion modules (DFM), and a multiscale change decoder (MCD).

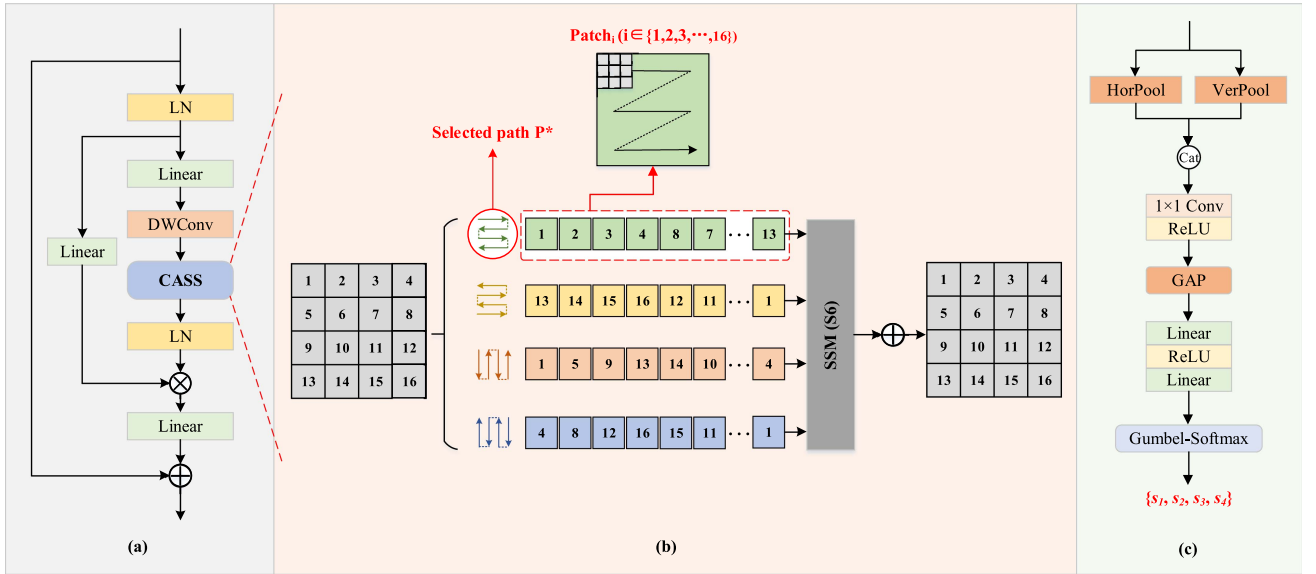


Fig. 2. Each stage of the encoder contains a CASS Block (a), and its core channel adaptive state-space scan strategy (b) selects the key channel through a lightweight scanning channel attention scorer (c), and enhances the local features within the patched feature maps. (a) CASS Block. (b) CASS Strategy. (c) LSCA scorer.

we propose the CASS encoder, which enhances the scanning strategy of the Mamba architecture.

The overall architecture of CASS encoder is built upon VMamba and is divided into four stages. As shown in Fig. 2(a), the input is divided into two information streams. The main stream passes through a LN layer, a linear layer, and a depthwise convolution (DWConv). After being processed

by the CASS strategy, the output is passed through another LN layer. Then, this output is fused with a linearly transformed version of the input feature using the Hadamard product, producing the output of the main stream. The second information stream is integrated into the main stream output via a residual connection to obtain final outputs X_i and Y_i .

The CASS strategy serves as the central element of the encoder and is designed to improve the scanning approach used in the Mamba architecture. As shown in Fig. 2(b), CASS unfolds the input patches into feature sequences along four different paths. To mitigate the spatial discontinuities of feature information that arise at row or column transitions in the commonly used Z-shaped scanning strategy, CASS employs an S-shaped scanning path to maintain local boundary continuity. A lightweight scanning channel attention (LSCA) scorer evaluates the importance of each direction and outputs four normalized scores $\{s_1, s_2, s_3, s_4\}$. These scores are produced by global pooling and bottleneck projection, followed by a Gumbel-Softmax operation, which encourages them to approximate a one-hot distribution. Selected path P^* indicates the selection of the highest-scoring scanning direction, formally defined as follows:

$$P^* = \operatorname{argmax} \{s_1, s_2, s_3, s_4\}. \quad (7)$$

Before initiating scanning along P^* , a 3×3 Conv kernel is utilized to extract local features within each patch to enhance edges, textures, and detailed information. The other three directions are still retained but only flattened into sequences. The feature sequences from the four directions are stacked along a new dimension and fed into the S6 block. The S6 block independently processes each sequence, performing global modeling in specific directions. Finally, all output features are merged to obtain a new feature map that includes global modeling information from all four directions.

The detailed architecture of LSCA scorer is illustrated in Fig. 2(c), which obtains the weights of four scanning paths with a minimal number of parameters. Given a feature map, two one-dimensional pooling layers are utilized to explicitly capture features in horizontal and vertical directions, providing direction-sensitive inputs for subsequent path scoring. The results from different directions are expanded and concatenated, and then input into a global average pooling (GAP) layer, which retains only channel-wise features and focuses on the existence of features rather than their specific positions.

Then, a bottleneck layer maps the channel features output from GAP to weights for the four scan paths. The designed bottleneck layer improves feature extraction efficiency while simultaneously minimizing the number of parameters. Its structure is defined as follows:

$$\text{Bottleneck}(X) = \text{Linear}(\text{ReLU}(\text{Linear}(X))). \quad (8)$$

Due to the requirement for the LSCA scorer to provide an approximate one-hot output for determining the key scanning direction, a Gumbel-Softmax function is used to generate differentiable weights, ensuring gradient flow during training. It introduces two parameters, noise and temperature, based on the Softmax function, which can be described as follows:

$$\begin{cases} y_{i_Softmax} = \exp(z_i) / \sum_j \exp(z_j) \\ y_{i_Gumbel} = \exp((z_i + g_i) / \tau) / \sum_j \exp((z_j + g_j) / \tau) \end{cases} \quad (9)$$

where z is the input logits vector, g represents the Gumbel noise, and τ represents the temperature. The traditional Softmax function depends entirely on the input z , while Gumbel-Softmax

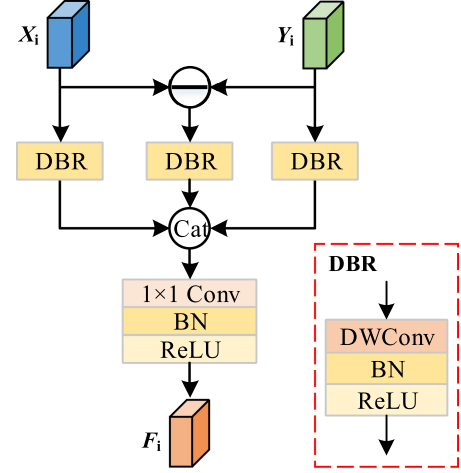


Fig. 3. Architecture of DFM, which employs three branches to fuse bitemporal features and difference information.

introduces noise to make the sampling results closer to the true discrete distribution, preventing them from getting stuck in local optima. In addition, by using a larger temperature parameter τ in the early stages of training to control the output results to approach a uniform distribution, the model is encouraged to explore all scanning directions. Then, an annealing strategy is used to reduce the temperature parameter τ and gradually discretize the sampling, ultimately obtaining an approximate one-hot output, thereby determining the most important direction containing key feature information. Since the LSCA scorer relies solely on average pooling layers, which contain no trainable parameters, and learns the mapping from feature channels to scan paths via a simple linear layer, the computational cost is very low.

The CASS encoder employs a lightweight scorer to select the most crucial scan channel and enhance local features of the patched feature maps. It improves the network's capacity to identify key features without introducing significant additional parameters.

D. Difference Fusion Module

To additional boost the network's capacity to capture local features and detect subtle changes, we introduce a DFM after each stage of the encoder, as shown in Fig. 3. The bitemporal features X_i and Y_i output from each encoder stage are used as inputs to the DFM, which processes the feature information through three branches. Each of the three branches in DFM processes feature information through a combination of a DWConv, a batch normalization (BN) layer, and a rectified linear unit (ReLU). Specifically, the bitemporal feature branch employs a 3×3 DWConv to preserve fine-grained edge information. In contrast, the difference branch adopts a larger 5×5 kernel, since direct differencing of bitemporal images may introduce pseudochange noise due to illumination variations and registration errors, and the enlarged receptive field facilitates the suppression of spurious changes and the continuity of boundaries. The outputs from the three branches are concatenated along the channel dimension, succeeded by a 1×1 Conv to adjust the number

of channels. A BN layer and a ReLU layer are then applied to generate the output F_i . The complete DFM operation can be formulated as follows:

$$\begin{cases} f_x = \text{ReLU}(\text{BN}(\text{DWConv}_{3 \times 3}(X_i))) \\ f_y = \text{ReLU}(\text{BN}(\text{DWConv}_{3 \times 3}(Y_i))) \\ f_{diff} = \text{ReLU}(\text{BN}(\text{DWConv}_{5 \times 5}(X_i \ominus Y_i))) \\ f_i = f_x \oplus f_y \oplus f_{diff} \\ F_i = \text{ReLU}(\text{BN}(\text{Conv}_{1 \times 1}(f_i))) \end{cases} \quad (10)$$

where $\ominus$ represents element-wise subtraction, and $\oplus$ represents channel-wise concatenation.

While preserving the local features of bitemporal images at different scales, F_i also integrates difference information, thus boosting the network's capacity to capture detailed changes.

E. Multiscale Change Decoder

To reconstruct the final change map, we design a simplified yet effective decoder that exclusively relies on the multiscale difference fusion features $\{F_1, F_2, F_3, F_4\}$ produced by DFM. This design makes full use of the bitemporal and difference information embedded in F_i , while eliminating the redundant usage of original bitemporal features X_i and Y_i , thereby avoiding the propagation of irrelevant background information. The decoding process adopts a hierarchical upsampling and fusion strategy. As illustrated in Fig. 1, starting from the deepest feature F_4 , we progressively upsample and fuse it with shallower features. At each stage i , the decoding operation is formulated as follows:

$$F'_i = \text{CBR}(\text{SE}(\text{Up}(F'_{i+1} \oplus F_i))), \quad i = 3, 2, 1. \quad (11)$$

In this formula, $\oplus$ denotes channel-wise concatenation. $\text{Up}(\cdot)$ denotes a $2 \times$ upsampling operation implemented with DySample, a lightweight and efficient upsampler. $\text{SE}(\cdot)$ denotes a lightweight attention module (SE Block), which is used to adaptively enhance features of key channels while suppressing irrelevant information. $\text{CBR}(\cdot)$ is a composite operation of 1×1 Conv, BN, and ReLU, used for channel alignment.

Finally, the shallowest feature F'_1 is fed into a predictor head that consists of an MLP and two successive upsampling stages. The MLP introduces nonlinear feature interactions, and the upsampling stages gradually restores the spatial resolution to the input size. A final 1×1 Conv followed by a Softmax layer produces the binary change mask.

IV. EXPERIMENTS AND ANALYSIS

A. Datasets

In this study, four widely used publicly available datasets are selected: the LEVIR-CD dataset [3], the WHU-CD dataset [54], the SYSU-CD dataset [55], and the CLCD dataset [29], which are detailed as follows.

- 1) **LEVIR-CD**: LEVIR-CD [3] is a publicly accessible, large-scale RSCD dataset comprising 637 pairs of very high-resolution (VHR) images, each with a spatial resolution of 0.5 m per pixel and dimensions of 1024×1024 pixels. For preprocessing, the original images are divided into nonoverlapping patches of size 256×256 . Based on a 7:1:2 ratio, the dataset is partitioned into 7120 images for

training, 1024 images for validation and 2048 images for testing.

- 2) **WHU-CD**: WHU-CD [54], derived from the broader WHU building dataset, is specifically designed for building CD tasks. It comprises a pair of large-scale RS images of New Zealand, each with dimensions of $32\,507 \times 15\,354$ pixels and a spatial resolution of 0.2 meters per pixel, captured in April 2012 and April 2016, respectively. We partition the images into 256×256 patches, yielding 7620 individual samples. Based on a 7:1.5:1.5 ratio, the dataset is partitioned into 5334 images for training, and 1143 images each for validation and testing.
- 3) **SYSU-CD**: SYSU-CD [55] contains 20 000 pairs of aerial images collected over Hong Kong between 2007 and 2014, with each image having a spatial resolution of 0.5 meters per pixel and dimensions of 256×256 pixels. This dataset emphasizes change detection in urban and coastal environments, characterized by dense high-rise construction and ongoing infrastructure development. Based on a 6:2:2 ratio, the dataset is partitioned into 12 000 images for training, and 4000 images each for validation and testing.
- 4) **CLCD**: CLCD [29] is a dataset designed for detecting changes in cultivated land, comprising 600 pairs of RS images with dimensions of 512×512 pixels and spatial resolutions ranging from 0.5 to 2 m. The dataset captures various types of land surface changes. For preprocessing, the images are divided into 256×256 patches, yielding a total of 2400 samples. Based on a 6:2:2 ratio, the dataset is partitioned into 1440 images for training, and 480 images each for validation and testing.

B. Experimental Setup and Evaluation Metrics

The SCAM model is implemented and trained via the PyTorch framework on a single NVIDIA RTX 3080Ti GPU, which features 12 GB of memory. All four datasets are preprocessed by cropping the images into 256×256 patches. To boost data variety, we apply augmentation methods like flipping, transposition, and bitemporal image swapping. During training, the network is optimized using the AdamW optimizer [56], with an initial learning rate of 1×10^{-3} and a weight decay of 5×10^{-3} . The batch size is set to 8, and training is conducted 80 000 iterations. Model performance is assessed on the respective test sets of each dataset.

The encoder of the proposed SCAM architecture consists of four stages, where the size of bitemporal images feature maps is progressively reduced to $\{1/4, 1/8, 1/16, 1/32\}$ of the input dimensions. Correspondingly, the quantity of feature channels is set to $\{64, 128, 192, 256\}$. Downsampling within each stage is performed using linear interpolation.

Considering the issue of class imbalance in RSCD tasks, we utilize a hybrid loss function combining balanced cross-entropy (BCE) loss [57], dice loss [58], and focal loss [59] to guide the learning process of the change mask. The loss function is formulated as follows:

$$\mathcal{L} = \alpha \cdot \mathcal{L}_{\text{BCE}} + \beta \cdot \mathcal{L}_{\text{Dice}} + \gamma \cdot \mathcal{L}_{\text{Focal}} \quad (12)$$

TABLE I
COMPARISONS RESULTS ON LEVIR-CD AND WHU-CD DATASETS

Type	Method	LEVIR-CD					WHU-CD				
		Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA
<i>C</i>	FC-Siam-Conc [18]	85.93	86.23	86.08	74.98	98.59	88.67	78.32	83.17	69.62	98.43
	SNUNet [21]	86.67	89.18	87.91	81.43	99.01	85.87	82.86	84.08	73.78	98.75
	WNet [48]	92.15	88.97	90.53	82.88	99.04	90.35	92.19	91.26	83.87	99.31
	SEIFNet [50]	91.35	89.64	90.49	83.15	99.06	87.01	85.77	86.39	76.04	98.90
<i>T</i>	BIT [28]	91.52	86.13	88.74	80.43	98.91	86.11	92.02	88.97	79.12	98.92
	ChangeFormer [27]	84.12	88.01	86.02	76.82	98.70	82.63	87.23	84.87	75.43	98.69
	TransUNetCD [51]	88.32	90.14	89.22	80.18	98.94	88.61	91.73	90.14	81.41	98.99
<i>M</i>	RS-Mamba [43]	89.62	90.17	89.89	82.36	98.99	92.73	90.45	91.59	85.33	99.27
	ChangeMamba [52]	89.26	91.62	90.42	81.98	99.02	92.52	89.62	91.05	83.51	99.21
	CD-Lamba [44]	89.98	91.06	90.52	82.01	99.05	93.05	92.14	92.59	86.11	99.33
	SCAM	90.79	91.23	91.01	83.28	99.11	93.18	93.09	93.14	86.83	99.41

Note: C, T, M represent CNN-based methods, transformer-based methods and Mamba-based methods, respectively. All scores are in percentage. The top two results are highlighted in red and green.

TABLE II
COMPARISONS RESULTS ON SYSU-CD AND CLCD DATASETS

Type	Method	SYSU-CD					CLCD				
		Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA
<i>C</i>	FC-Siam-Conc [18]	75.82	74.71	75.26	62.94	89.63	66.24	53.72	59.33	40.72	94.55
	SNUNet [21]	80.52	75.17	77.75	64.15	90.02	74.31	60.23	66.53	48.41	95.15
	WNet [48]	81.21	80.13	80.67	67.35	90.99	75.68	71.39	73.47	59.57	95.71
	SEIFNet [50]	84.81	79.98	82.32	69.96	91.90	76.63	72.61	74.57	60.18	95.89
<i>T</i>	BIT [28]	82.61	78.45	80.48	65.84	90.93	72.82	64.70	68.52	53.83	95.39
	ChangeFormer [27]	77.92	80.13	79.01	64.82	90.35	69.32	62.94	65.98	51.72	95.02
	TransUNetCD [51]	80.24	79.54	79.89	66.83	91.07	73.24	64.71	68.71	54.83	95.34
<i>M</i>	RS-Mamba [43]	81.93	79.02	80.44	66.92	91.12	76.25	77.12	76.68	61.51	96.12
	ChangeMamba [52]	82.56	79.83	81.17	67.11	91.36	80.41	74.92	77.57	65.14	96.84
	CD-Lamba [44]	84.25	80.81	82.50	70.38	91.70	79.02	77.05	78.02	63.97	96.77
	SCAM	84.91	82.33	83.60	70.81	92.25	79.26	78.43	78.84	64.92	96.89

Note: C, T, M represent CNN-based methods, transformer-based methods, and Mamba-based methods, respectively. All scores are in percentage. The top two results are highlighted in red and green.

where the weight coefficients α , β , and γ are set to 0.3, 0.3, and 0.4, respectively.

To assess the effectiveness of the proposed model, we adopt five widely used evaluation metrics: precision (Pre), recall (Rec), overall accuracy (OA), F1 score (F1), and intersection over union (IoU). Precision quantifies the ratio of correctly predicted positive pixels to total number of pixels predicted as positive. Recall measures the proportion of actual positive pixels that are correctly classified. OA indicates the proportion of all pixels that are accurately predicted. The F1 score represents the harmonic mean of precision and recall, serving as a comprehensive measure of model performance. IoU calculates the degree of overlap between the predicted positive region and the corresponding ground truth. The formulas for the above evaluation metrics are defined as follows:

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) \quad (13)$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) \quad (14)$$

$$\text{F1} = (2 \times \text{Pre} \times \text{Recall}) / (\text{Pre} + \text{Recall}) \quad (15)$$

$$\text{OA} = (\text{TP} + \text{TN}) / (\text{TP} + \text{FP} + \text{TN} + \text{FN}) \quad (16)$$

$$\text{IoU} = \text{TP} / (\text{TP} + \text{FN} + \text{FP}) \quad (17)$$

where TP, FP, TN, and FN denote the number of true positives, false positives, true negatives, and false negatives, respectively.

C. Performance Comparison

To verify the effectiveness of the proposed method, we conduct comparisons with several state-of-the-art CD methods. The CNN-based methods include FC-Siam-Conc [18], SNUNet [21], WNet [48], and SEIFNet [50]. The transformer-based methods include BIT [28], ChangeFormer [27], and TransUNetCD [51]. The Mamba-based methods include RS-Mamba [43], ChangeMamba [52], and CD-Lamba [44]. We perform experiments across four datasets: LEVIR-CD, WHU-CD, SYSU-CD, and CLCD. The quantitative comparison results are illustrated in Tables I and II. All evaluation outcomes are presented as percentages (%). The top-performing results are marked in red, while the second-best are indicated in green. Visual comparisons across the four datasets are illustrated in Figs. 4–7, where different colors represent the detection results:

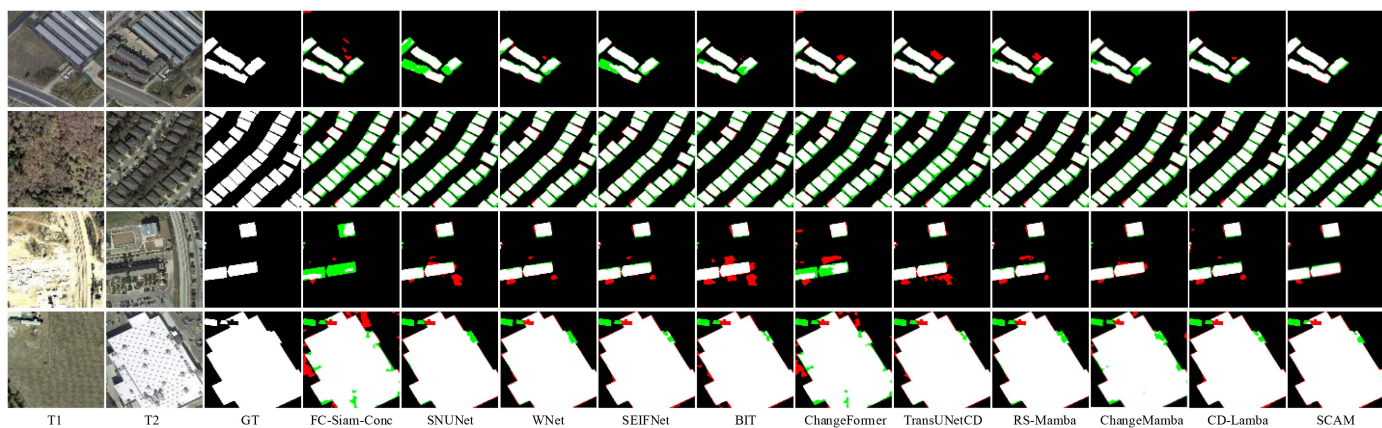


Fig. 4. Visualization results of LEVIR-CD, where white indicates true positive, black indicates true negative, red indicates false positive, and green indicates false negative.

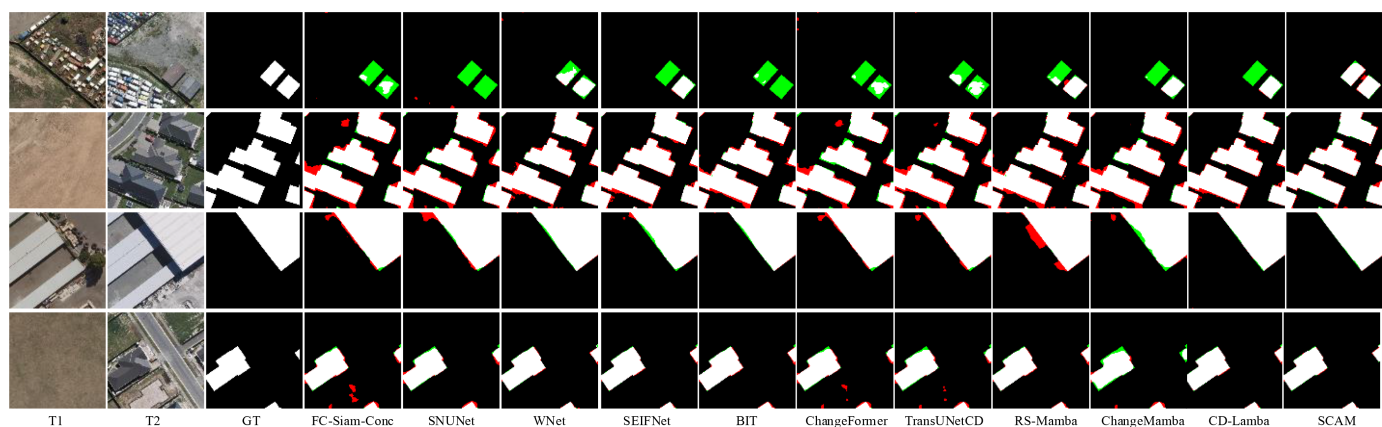


Fig. 5. Visualization results of WHU-CD, where white indicates true positive, black indicates true negative, red indicates false positive, and green indicates false negative.

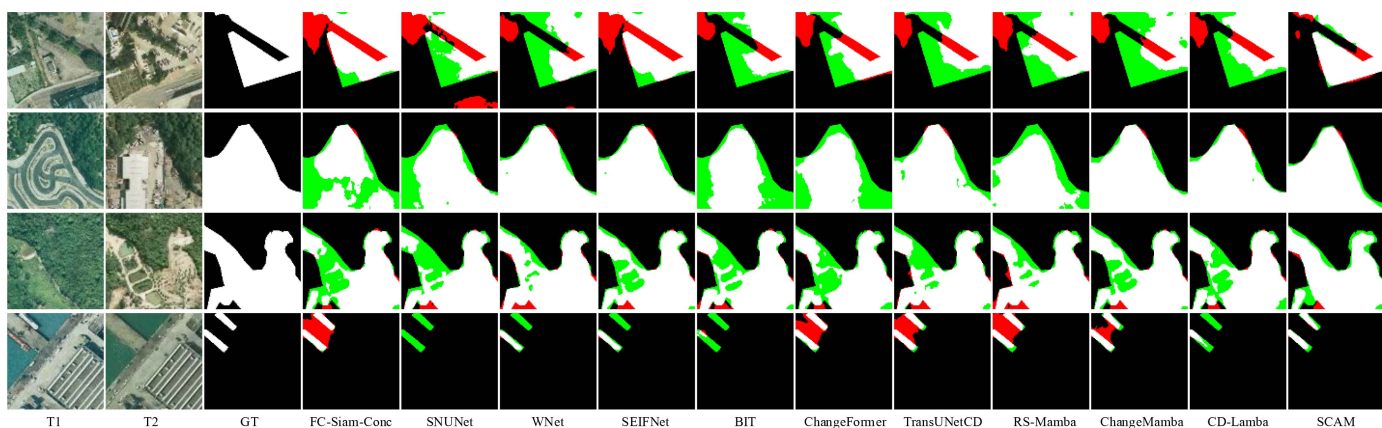


Fig. 6. Visualization results of SYSU-CD, where white indicates true positive, black indicates true negative, red indicates false positive, and green indicates false negative.

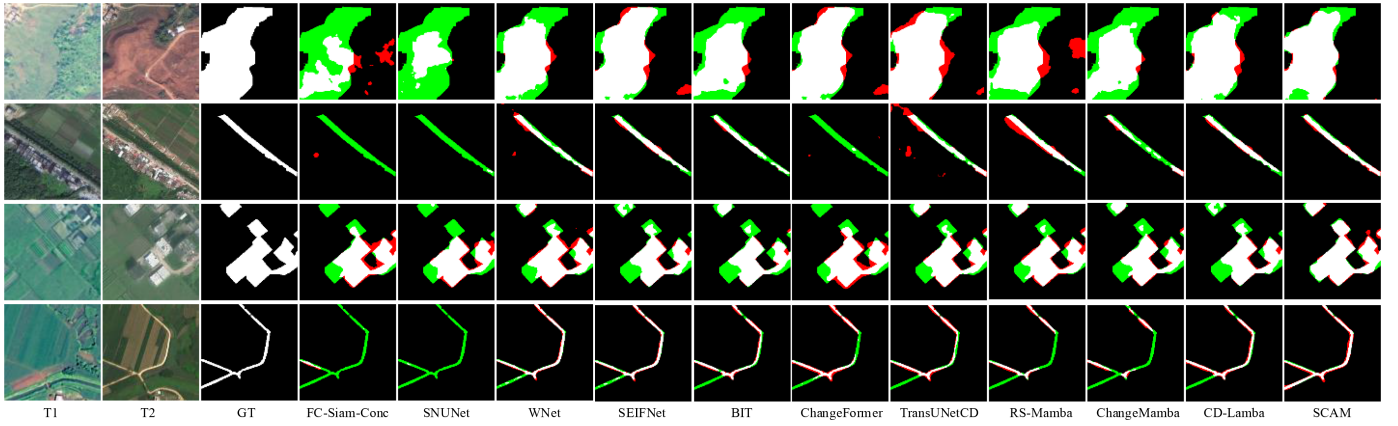


Fig. 7. Visualization results of CLCD, where white indicates true positive, black indicates true negative, red indicates false positive, and green indicates false negative.

TABLE III
EFFICIENCY COMPARISON RESULT

Type	Method	Params(M)	GFLOPs
C	FC-Siam-Conc	1.54	5.32
	SNUNet	10.20	44.38
	WNet	43.07	19.20
	SEIFNet-R	27.91	8.37
T	BIT-18_S4	3.55	4.35
	ChangeFormer	41.03	202.87
	TransUNetCD	95.49	27.23
M	RS-Mamba	27.90	15.70
	ChangeMamba_Small	49.94	114.82
	CD-Lamba	28.74	15.26
	SCAM	25.75	21.13

Note: C, T, M represent CNN-based methods, transformer-based methods, and Mamba-based methods, respectively.

white, black, red, and green denote TP, TN, FP, and FN, respectively. We also compare the efficiency of all models, as shown in Table III.

1). *Analysis of LEVIR-CD and WHU-CD*: LEVIR-CD and WHU-CD are building-oriented datasets that primarily record structural changes in buildings. Due to the relatively single of the samples, we analyze the results of these two datasets together. According to Table I, it is evident that the proposed SCAM outperforms the other methods in overall performance.

Compared with the recently proposed CD-Lamba, SCAM improves the F1 score by 0.49% and 0.55% on LEVIR-CD and WHU-CD, respectively. Specifically, within CNN-based methods, WNet and SEIFNet exhibit notable superiority, attributed respectively to the strong modeling capability of the dual-Siamese branch encoder and the flexible feature extraction approach. Transformer-based methods, with their powerful global modeling capabilities, effectively address the limited receptive field problem of CNNs and perform well overall in CD tasks. For example, BIT leverages the Transformer architecture and semantic tokens to integrate change areas between bitemporal images, thereby improving the efficiency and accuracy of change detection, highlighting the unique advantage of

transformers in this field. Mamba-based approaches introduced recently have generally achieved superior performance compared to both CNNs and transformer, benefiting from Mamba's ability to effectively capture and integrate global features. The proposed SCAM attains the highest F1 score on both datasets, with 91.01% on LEVIR-CD and 93.14% on WHU-CD. This clearly demonstrates that SCAM enhances the perception of important features and change details, achieving an ideal balance between precision and recall. Moreover, SCAM also achieves the highest IoU and OA, showing significant advantages over other Mamba-based methods.

To additionally demonstrate the performance of SCAM, we visualize the detection results of several representative CD methods on the LEVIR-CD and WHU-CD test sets. As illustrated in Figs. 4 and 5, it is evident that CNN-based methods tend to produce more false positives and false negatives. The performance of transformer-based methods shows noticeable improvement. However, these methods still struggle to capture change details. In contrast, the proposed SCAM demonstrates the best overall performance. Specifically, as illustrated in the first image group of Fig. 4 and the first image group of Fig. 5, SCAM effectively captures small change regions that are difficult to detect with other methods and provides more complete change shapes. In addition, as illustrated in the third image group of Fig. 5, the boundaries of the change areas detected by SCAM are noticeably clearer and more accurate. These results validate that our method can effectively enhance local detail features and improve the model's ability to perceive details in changing areas.

2). *Analysis of SYSU-CD and CLCD*: The SYSU-CD and CLCD datasets cover various types of land surface changes and involve relatively complex scenes. Therefore, we analyze the results on these two datasets together.

According to Table II, it is evident that the proposed SCAM achieves the highest F1 score on both datasets, outperforming the second-best methods by 1.10% and 0.82%, respectively. These improvements are significantly larger than those observed on the previous two datasets, demonstrating the enhanced performance of our approach under complex conditions. Specifically, CNN-based methods perform poorly on complex datasets because they lack the

capacity to capture global contextual information. Notably, transformer-based methods achieve performance close to that of Mamba-based approaches on the SYSU-CD dataset. Nevertheless, there is a significant gap in their performance on the CLCD dataset. We hypothesize that this issue is influenced by both the size and quality of the datasets. Transformer's self-attention mechanism effectively models long-range dependencies, but it requires large amounts of data to effectively learn complex spatial features and change patterns. As a result, their performance on the large-scale, high-quality SYSU-CD dataset is relatively strong. In contrast, the CLCD dataset is smaller in scale and contains pseudochange noise caused by lighting differences, which hinders the effectiveness of Transformer. In such cases, the self-attention mechanism may even amplify global noise.

In comparison, Mamba-based methods exhibit clear advantages on both datasets. For instance, ChangeMamba obtains the highest precision of 80.41% and the highest IoU of 65.14% on the CLCD dataset, surpassing all other methods. Nevertheless, SCAM attains the highest recall, F1 score and OA across both datasets, highlighting outstanding performance compared to other models.

In addition, we visualize the detection results on the SYSU-CD and CLCD test sets. As depicted in Figs. 6 and 7, it is clear that SCAM produces fewer false positive and false negative regions across various change scenarios. Specifically, as observed in the first image group of Fig. 6, SCAM demonstrates strong robustness in vegetation-covered areas with color and lighting differences, resulting in fewer false positives. From the third image group in Fig. 6, we can observe that SCAM effectively captures the elements within the change region, avoiding false negatives in that area. On the CLCD dataset, methods such as SNUNet and FC-Siam-Conc produce a large number of false negatives and almost completely fail to detect road changes, highlighting the limitations of CNN-based models in complex CD tasks. In the second and fourth image groups of Fig. 7, although SEIFNet, WNet and CD-Lamba can also detect parts of the road changes, only SCAM generates a more complete change mask while minimizing false positive errors. These observations further validate that our method can significantly enhance model performance in complex scenes.

3) *Analysis of Efficiency*: To evaluate model efficiency, we compare each model in terms of the parameters (Params) and giga floating-point operations per second (GFLOPs). Version labels are assigned to models that contain multiple variants. The input bitemporal images are of size $256 \times 256 \times 3$.

As illustrated in Table III, FC-Siam-Conc demonstrates a clear advantage in terms of model efficiency, but this comes at the cost of poor model performance. Transformer-based methods, such as ChangeFormer, incur substantial computational overhead due to quadratic complexity. Among Mamba-based methods, SCAM has the smallest number of parameters (25.75M), with GFLOPs slightly higher than those of RS-Mamba and CD-Mamba. This indicates that SCAM achieves an effective balance between parameter count and computational cost, making it well-suited for deployment on resource-constrained devices.

TABLE IV
SCAN STRATEGY ABLATION STUDY ON LEVIR-CD

Strategy	Pre	Rec	F1	IoU	OA
SS2D	89.13	90.81	89.96	81.72	98.96
Local-Scan	88.23	92.15	90.15	81.26	98.93
OSS	90.52	90.28	90.40	82.87	99.01
CT-LASS	89.92	91.03	90.47	82.79	99.03
CASS_Z	90.29	91.15	90.72	83.07	99.08
CASS_S	90.79	91.23	91.01	83.28	99.11

Note: Highest scores are in bold.

D. Ablation Studies

To assess the contributions of each strategy and module in SCAM, we perform a series of ablation experiments across the LEVIR-CD dataset. The details are as follows.

1) *Effects of CASS Strategy*: To evaluate the efficacy of the CASS strategy, we design five comparison groups.

Specifically, we retain all modules and parameters in SCAM while replacing CASS with five different scan strategies. These include the SS2D from VMamba, the Local-Scan from Local-Mamba, the OSS from RS-Mamba, and the CT-LASS from CD-Lamba. To validate the effectiveness of the S-shaped scanning path (CASS_S) in optimizing local boundaries, we additionally establish a control group employing the CASS strategy with a Z-shaped scanning path (CASS_Z).

As indicated in Table IV, all the other strategies achieves better performance than SS2D, which performs a flattened sequential scan across the image. Among them, Local-Scan achieves the highest recall of 92.15%. However, the proposed CASS_S strategy attains the highest precision, F1 score, IoU, and OA, demonstrating the best overall performance. As illustrated in Fig. 8, we compare the visual results of different scan strategies across the LEVIR-CD test set. In the second image group, it is clearly observed that SS2D produces numerous false positives in the continuous change region at the top. While Local-Scan shows improved performance in this challenging area, there are still many misdetections near the overall boundaries. This may be because the Local-Scan strategy breaks the overall correlation of the image, and the discontinuous scanning at the boundaries leads to false detection of edge pixels. OSS effectively eliminates most false positives around the outer region of the change area at the top.

However, it still produces a significant number of errors in the narrow spaces between newly added buildings. We speculate that although OSS enhances robustness through global scanning, it still tends to capture macrolevel changes, making it less sensitive to local features. CT-LASS and CASS_Z strategies substantially alleviate the interference of pseudochange regions at the top through adaptive enhancement of local features. Nevertheless, owing to the discontinuities introduced by the Z-shaped scanning path, numerous false positives persist along the upper and lower boundaries of the corresponding change region. In contrast, the CASS_S strategy exhibits the strongest robustness in the top change region of the second image group, effectively eliminating the interference of pseudochange noise.

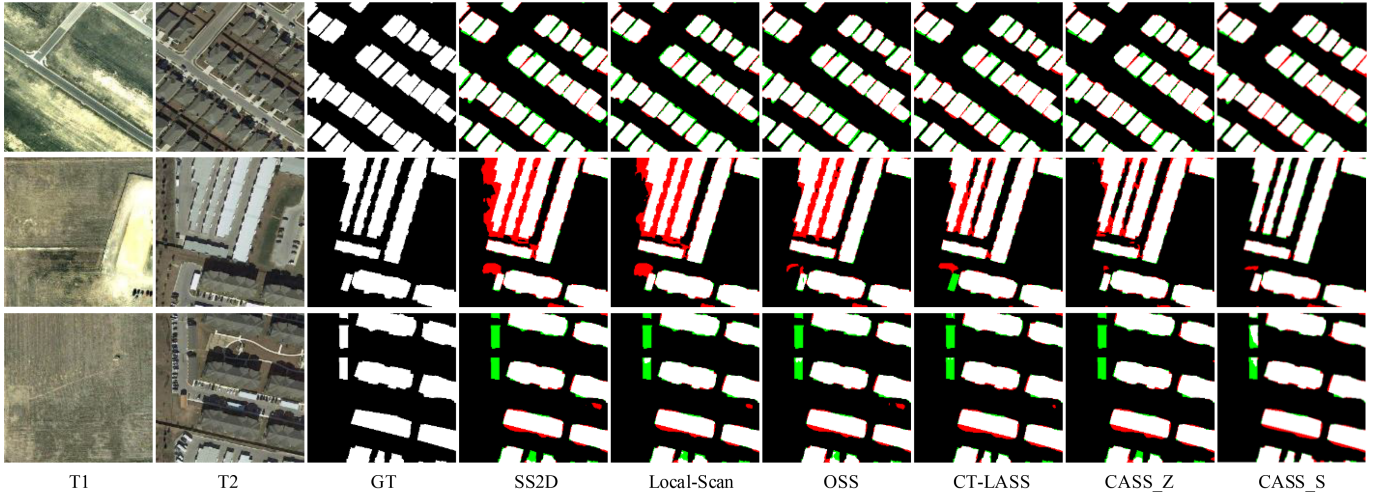


Fig. 8. Visualization results of scan strategy ablation study on LEVIR-CD, where white indicates true positive, black indicates true negative, red indicates false positive, and green indicates false negative.

TABLE V
ABLATION STUDY ON DFM AND MCD ON LEVIR-CD

Baseline	DFM	MCD	F1	IoU	OA
✓	×	×	84.91	76.32	98.66
✓	✓	×	88.77	81.89	98.90
✓	✓	✓	91.01	83.28	99.11

Note: Highest scores are in bold.

By assessing key scanning channels and reinforcing local features within their patched feature maps, together with the adoption of the S-shaped scanning path, the proposed CASS strategy improves both the perception of critical features and the modeling of local change boundaries. These findings substantiate the efficacy of CASS strategy in balancing global context with local detail preservation.

2) *Effects of DFM and MCD*: To assess the impact of the proposed DFM and MCD, we perform three groups of ablation studies. The baseline model is composed of the CASS encoder and a simple four-layer upsampling decoder. As shown in Table V, the first group includes only the baseline architecture. In the second group, the DFM is added, and its multiscale features are integrated into different stages of the decoder. In the third group, based on the second setup, the upsampling decoder is replaced with the proposed MCD. The outcomes of experiments indicate that adding the DFM alone results in significant improvements of 3.86% in F1 score and 5.57% in IoU compared to baseline, demonstrating that DFM effectively boosts the network's capacity to detect change details by integrating difference information with local features of bitemporal images. When DFM is combined with MCD, the model is additionally optimized, achieving the best performance. This indicates that the proposed MCD is well-suited to deal with the multiscale information captured by DFM, enabling more effective decoding and change localization.

To further investigate the contribution of DFM at different encoder stages to feature optimization, as well as the

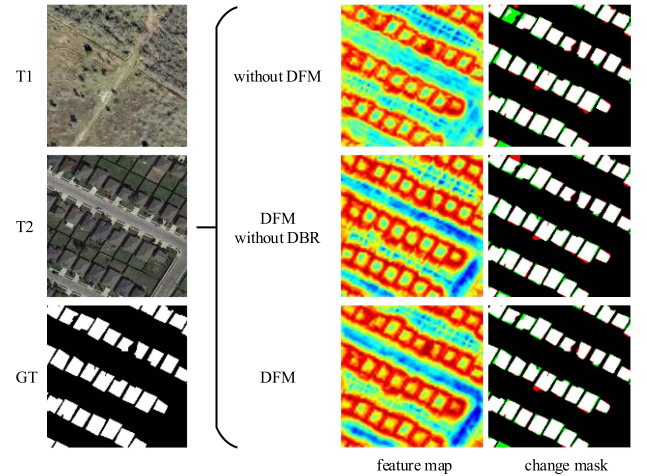


Fig. 9. Visualization of feature maps and prediction of change masks on LEVIR-CD test set. Red represents high response value areas, and blue represents low response value areas.

effectiveness of its internal structure, we conduct six groups of ablation studies. As shown in Table VI, the experimental results indicate that the model performance consistently improves as DFM is deployed in more encoder stages. It is noteworthy that introducing DFM in the first two stages yields more significant improvements in the F1 score, suggesting that DFM plays a crucial role in enhancing fine-grained difference information within shallow, large-scale feature maps. In addition, we design a variant of DFM to explore the advantages of DFM's internal structure in local detail modeling and noise suppression. In this variant, the DBR operations of the three branches are removed, rendering its structure analogous to the DEM [48] in WNet. Fig. 9 visually demonstrates the feature optimization effect of DFM. Without using DFM, the extracted change feature contours are relatively blurry, resulting in more false positive and false negative errors in the edge regions of the change mask inferred by the model. In contrast, after deploying

TABLE VI
ABLATION STUDY ON DFM AT DIFFERENT STAGES ON LEVIR-CD

Stage	I	II	III	IV	Pre	Rec	F1	IoU	OA
DFM	×	×	×	×	87.92	89.64	88.77	81.79	98.74
	✓	×	×	×	89.32	90.31	89.81	82.77	98.93
	✓	✓	×	×	90.61	90.35	90.48	82.88	99.02
	✓	✓	✓	×	90.68	90.89	90.78	83.11	99.06
	✓	✓	✓	✓	90.79	91.23	91.01	83.28	99.11
DFM without DBR	✓	✓	✓	✓	90.63	91.20	90.91	83.17	99.10

Note: Highest scores are in bold.

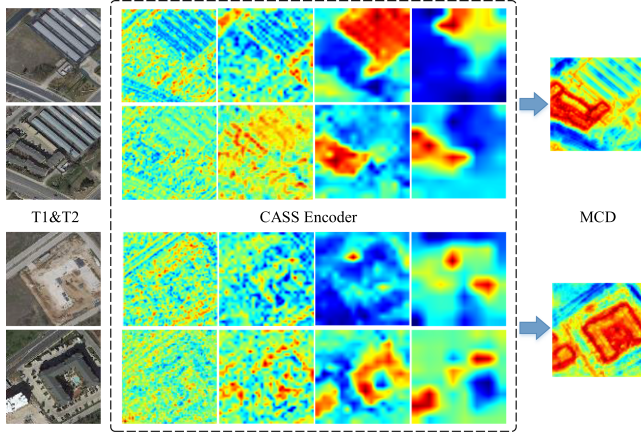


Fig. 10. Visualization results of features at different stages on LEVIR-CD test set. Red represents high response value areas, and blue represents low response value areas.

DFM in four encoder stages, the feature visualization results improve significantly and the feature boundaries become clearer. Although the DFM variant can also significantly optimize feature representations, its performance is inferior to that of the original DFM due to limited local modeling capability and susceptibility to pseudochange noise. Visualization results further reveal that the variant introduces more noisy artifacts along change boundaries, thereby generating additional false positives in the prediction maps. This leads to a notable decline in precision and IoU, which is consistent with the quantitative results reported in Table VI. The ablation study results validate that the proposed DFM can effectively enhance local difference information while suppressing pseudochange noise.

3) *Feature Visualization*: We further visualize the feature activation maps generated by the encoder and decoder to confirm the efficacy of the model. As shown in Fig. 10, the four stages of the CASS encoder capture information in the bitemporal images from shallow to deep. The shallow layers extract edge, texture, and structural details, while deeper layers encode semantic features. The MCD processes the multiscale features and produces feature maps with clearly defined regions of interest. It can be observed that the edges of the change regions exhibit strong activation responses, validating that the proposed SCAM, through the CASS strategy, effectively enhances the perception of important features and enables precise localization of change

areas. Furthermore, the combination of DFM and MCD facilitates the fusion of global and local information, leading in clearer and more complete boundaries and structures of the detected change areas.

V. CONCLUSION

This article presents a comprehensive exploration of applying Mamba-based models in the RSCD domain. We introduce a new SCAM to tackle the issue of Mamba-based methods being limited by fixed scanning strategies and unable to adaptively capture spatial contextual information and enhance local features. SCAM introduces the CASS strategy, a dynamic scanning mechanism capable of capturing and enhancing key features while maintaining global perception. In addition, we design DFM and MCD, which effectively integrate global and local features. Specifically, DFM embeds multiscale difference information into the global features of bitemporal images, thus boosting the network's capacity to detect change details. MCD receives the fused features produced by DFM as inputs, and it integrates multiscale information to infer the change mask.

The proposed SCAM achieves the peak F1 score across four benchmark datasets (LEVIR-CD, WHU-CD, SYSU-CD, and CLCD), demonstrating superior overall performance relative to other representative methods, particularly showing more pronounced advantages in complex scenes. However, we find that the advantage of SCAM in building datasets with relatively single samples is not significant, and it is prone to false negative errors when detecting small changing targets. In subsequent research, we aim to further improve the selective scanning strategy and local feature enhancement mechanisms for RSCD tasks, with a particular focus on boosting the network's sensitivity to minor change targets. We believe these advancements will contribute to the broader advancement and deployment of Mamba-based methods in the RSCD field.

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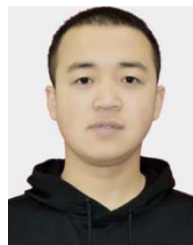
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